

# KALEY BRAUER

Harvard University, Center for Astrophysics | Harvard & Smithsonian  
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## EDUCATION

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| <b>Ph.D. in Physics</b>                                                                                                 | 2023 |
| Massachusetts Institute of Technology<br><i>The Legacy of the First Galaxies</i> , Advisor: Anna Frebel<br>GPA: 5.0/5.0 |      |
| <b>B.Sc. in Physics</b>                                                                                                 | 2017 |
| Brown University<br>University honors, departmental honors, GPA: 4.0/4.0                                                |      |

## ACADEMIC POSITIONS

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| <b>NSF Postdoctoral Fellow</b>                                      | 2023 – present |
| Center for Astrophysics   Harvard & Smithsonian                     |                |
| <b>US Department of Energy Graduate Fellow; Graduate Researcher</b> | 2017 – 2023    |
| Massachusetts Institute of Technology                               |                |
| <b>Visiting Research Fellow</b>                                     | Fall 2021      |
| Lawrence Berkeley National Laboratory                               |                |

## HONORS & AWARDS

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|--------------------------------------------------------------------------------------------|-------------|
| Research Corporation for Science Advancement Fellow                                        | 2025        |
| NSF Astronomy & Astrophysics Postdoctoral Fellowship (\$330K; ~8 awarded annually)         | 2023        |
| Spot Awards, MIT, <i>in recognition of community service to School of Science</i>          | 2020 & 2022 |
| DOE Computational Science Graduate Fellowship (~\$450K; ~25 awarded annually)              | 2018        |
| NSF Graduate Research Fellowship (~\$150,000)                                              | 2018        |
| MIT Whiteman Fellowship (~\$100K)                                                          | 2017        |
| R. Bruce Lindsay Award, Brown University, <i>given to senior for excellence in physics</i> | 2017        |
| Eva A. Mooar Prize, Brown University, <i>given to woman for academic excellence</i>        | 2017        |
| Sigma Xi Research Honor Society                                                            | 2017        |
| Karen T. Romer Undergraduate Teaching and Research Award, Brown University                 | 2015        |

## PUBLICATIONS

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### Refereed

Including 8 first- or second-author and 1 first-author refereed workshop paper

### Journal Articles

- Andersson, E. P., Rey, M. P., Yates, R. M., Read, J. I., Agertz, O., Ji, A. P., Mead, J., **Brauer, K.**, Mac Low, M. (2025). EDGE-INFERNO: The impact of chemical enrichment assumptions on the individual stars of a simulated faint dwarf galaxy. *submitted to The Astrophysical Journal*.

14. Hazlett, R., Mead, J., Visbal, E., Bryan, G., Mac Low, M., Kulkarni, M., Andersson, E. P., **Brauer, K.**, Wise, J. (2025). From Primordial Stars to Early Galaxies: A Semi-Analytic Model Calibrated with AEOS and Renaissance. *submitted to The Astrophysical Journal*. arXiv:2510.11629
13. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Andersson, E. P., Frebel, A., Emerick, A., Cote, B. (2025). AEOS is Mixing it Up: The (In)homogeneity of Metal Mixing Following Population III Star Formation. *submitted to The Astrophysical Journal*. arXiv:2509.13580
12. **Brauer, K.**, Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Emerick, A., Andersson, E. P., Frebel, A., Cote, B. (2025). AEOS: The Impact of Pop III Initial Mass Function and Star-by-Star Models in Galaxy Simulations. *The Astrophysical Journal*, 993, 2.
11. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Emerick, A., Andersson, E. P., Frebel, A., Cote, B. (2025). AEOS: Transport of metals from minihalos following Population III stellar feedback. *The Astrophysical Journal*, 980, 62.
10. **Brauer, K.**, Emerick, A., Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Cote, B., Andersson, E. P., Frebel, A. (2025). AEOS: Star-by-Star Cosmological Simulations of Early Chemical Enrichment and Galaxy Formation. *The Astrophysical Journal*, 980, 41.
9. Chiti, A., Mardini, M. K., Limberg, G., Frebel, A., Ji, A. P., Reggiani, H., Ferguson, P., Andales, H. D., **Brauer, K.**, Li, T. S., Simon, J. D. (2024). Signatures of Extragalactic First Stars in the Large Magellanic Cloud. *Nature Astronomy*, 8, 637647.
8. **Brauer, K.**, Andales, H., Ji, A. P., Mardini, M., Frebel, A., Gomez, F. A., & O’Shea, B. W. (2022). Possibilities and Limitations of Kinematically Identifying Stars from Accreted Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 937, 14.
7. Ji, A. P., Naidu, R., **Brauer, K.**, Ting, Y., & Simon, J. (2022). Chemical Abundances of the Typhon Stellar Stream. *Monthly Notices of the Royal Astronomical Society*, 519, 4467-4478.
6. Mardini, M., Frebel, A., Chiti, A., Meiron, Y., **Brauer, K.**, Ou, X. (2022). Characterization of the Metal Weak Thick Disk of the Milky Way. *The Astrophysical Journal*, 936, 78.
5. **Brauer, K.**, Ji, A. P., Drout, M., Frebel, A. (2021). Collapsar R-Process Yields Can Reproduce [Eu/Fe] Abundance Scatter in Metal-Poor Stars. *The Astrophysical Journal*, 915, 81.
4. Gull, M., Frebel, A., Hinojosa, K., Roederer, I. U., Ji, A. P., **Brauer, K.** (2021). R-process-rich stellar streams in the Milky Way. *The Astrophysical Journal*, 912, 52.
3. **Brauer, K.**, Ji, A. P., Frebel, A., Dooley, G. A., Gomez, F. A., & O’Shea, B. W. (2019). The Origin of r-process Enhanced Metal-Poor Halo Stars In Now-Destroyed Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 871, 247.
2. **Brauer, K.**, Vrtillek, S. D., Peris, C., & McCollough, M. (2018). Phase-resolved spectroscopy of the low-mass X-ray binary V801 Ara. *Monthly Notices of the Royal Astronomical Society*, 478, 4894-4904.

### Refereed Workshop Papers

1. **Brauer, K.**, Dash, A. P., Vyas, M. J., Salim, A. (2025). From Simulations to Surveys: Domain Adaptation for Galaxy Observations. Machine Learning and the Physical Sciences (ML4PS) Workshop at NeurIPS 2025. Retrieved from [https://github.com/kaleybrauer/galaxy\\_da](https://github.com/kaleybrauer/galaxy_da)

## Non-Refereed

2. **Brauer, K.** (2021). “I’ll Finish it This Week” And Other Lies. *arXiv April Fools Paper*. arXiv:2103.16574
1. **Brauer, K.**, Ji, A., Hattori, K., Escobar, S., & Frebel, A. (2019). Kinematics of highly r-process-enhanced halo stars: Evidence for origins in now-destroyed ultra-faint dwarf galaxies. *Proceedings of the International Astronomical Union*, 14(S353), 71-74.

## PRESENTATIONS

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### INVITED

|                                                              |                                    |
|--------------------------------------------------------------|------------------------------------|
| Talk at From Galaxies to Planets: An Elemental Journey       | Ringberg Castle, Germany; Feb 2026 |
| Colloquium Talk at Rensselaer Polytechnic Institute (RPI)    | Troy, New York, USA; Oct 2025      |
| Astronomy & Data Science Talk at Yale University             | New Haven, CT, USA; May 2025       |
| Astronomy Seminar Talk at Columbia University                | New York, NY, USA; Mar 2025        |
| Fundamental Physics Seminar Talk at Brown University         | Providence, RI, USA; Mar 2025      |
| Astronomy Seminar Talk at Lund University                    | Lund, Sweden; Mar 2025             |
| Astronomy Seminar Talk at American Museum of Natural History | New York, NY, USA; Oct 2024        |
| Nuclear Astrophysics Seminar Talk at IReNA                   | East Lansing, MI, USA; Sep 2024    |
| Colloquium Summer Talk at Harvard-Smithsonian CfA            | Cambridge, MA, USA; July 2023      |
| Talk at US Department of Energy CSGF Program Review          | Washington, DC, USA; July 2022     |
| High Performance Computing Talks at IHPCSS                   | Athens, Greece; June 2022          |
| Colloquium Talk at University of Melbourne                   | Melbourne, Australia; Oct 2021     |
| Seminar Talk at Computational Research in Boston and Beyond  | Boston, USA; Oct 2021              |
| Seminar Talk at Carnegie Observatories                       | Pasadena, CA, USA; Oct 2020        |

### CONTRIBUTED

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| Poster at NeurIPS 2025 ML4PS Workshop                          | San Diego, CA, USA; Dec 2025   |
| Talk at IAU Symposium 403 on Galactic Outskirts                | Cordoba, Spain; Oct 2025       |
| Poster at Tracing Cosmic Evolution with Clusters of Galaxies V | Sesto, Italy; July 2025        |
| Talk at Galactic Frontiers II at Dartmouth                     | Hanover, CT, USA; Jun 2025     |
| Talk at First Galaxies Meeting at Oxford                       | Oxford, UK; Apr 2025           |
| Talk at NSF Postdoctoral Fellows Symposium 2025                | Washington DC, USA; Jan 2025   |
| Talk at IAU Symposium 395 on Stellar Populations               | Paraty, Brazil; Nov 2024       |
| Talk at Astrophysical Origins of Carbon                        | Tokyo, Japan; Sep 2024         |
| Talk at First Stars VII at CCA                                 | NYC, NY, USA; May 2024         |
| Talk at Galaxies from Scratch 2024                             | Vienna, Austria; Feb 2024      |
| Talk at 243rd Meeting of the American Astronomical Society     | New Orleans, LA, USA; Jan 2024 |
| Talk at NSF Postdoctoral Fellows Symposium 2024                | New Orleans, LA, USA; Jan 2024 |
| Talk at Flatiron CCA Galactic Frontiers                        | NYC, NY, USA; July 2023        |

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| Talk at IAU Symposium 377 on Early Disk Galaxy Formation     | Kuala Lumpur, Malaysia; Feb 2023 |
| Talk at 241st Meeting of the American Astronomical Society   | Seattle, WA, USA; Jan 2023       |
| Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting   | South Bend, IN, USA; May 2022    |
| Talk at 2021 GALAH Science Meeting                           | Sydney, Australia; June 2021     |
| Talk at <i>Linking the Galactic and Extragalactic</i>        | Wollongong, Australia; Dec 2020  |
| Talk at 235th Meeting of the American Astronomical Society   | Honolulu, HI, USA; Jan 2020      |
| Talk at IAU Symposium 353 on Galactic Dynamics               | Shanghai, China; July 2019       |
| Poster at JINA-CEE Frontiers in Nuclear Astrophysics Meeting | East Lansing, MI, USA; May 2019  |
| Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting   | South Bend, IN, USA; May 2018    |
| Poster at 229th Meeting of the American Astronomical Society | Grapevine, TX, USA; Jan 2017     |
| Poster at 47th Meeting of the Division of Planetary Science  | Washington, DC, USA; Nov 2015    |
| Talk at SETI Institute's Summer Colloquium Series            | Mountain View, CA, USA; Aug 2015 |

## TECHNICAL SKILLS

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|-----------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Programming</b>                | Python, C/C++, Julia, Java, HTML/CSS, SQL                                                              |
| <b>High Performance Computing</b> | OpenMP, MPI, OpenACC, experience with the National Energy Research Scientific Computing Center (NERSC) |

## TEACHING

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|------------------------------------------------------------------|-------------------------|
| <b>8.03 Vibrations and Waves Teaching Assistant</b>              | Spring 2023             |
| <i>Physics Department, Massachusetts Institute of Technology</i> |                         |
| <b>8.S30 Stellar Archaeology Teaching Assistant</b>              | Fall 2022               |
| <i>Physics Department, Massachusetts Institute of Technology</i> |                         |
| <b>Cosmology Course Designer and Instructor</b>                  | Spring 2021             |
| <i>Spark, MIT Educational Studies Program</i>                    |                         |
| <b>Catalyst Computer Science Instructor</b>                      | Spring 2019             |
| <i>Citizen Schools and Mass STEM Hub</i>                         |                         |
| <b>PHYS0220/0270 Astronomy Teaching Assistant</b>                | Fall 2014 - Spring 2015 |
| <i>Physics Department, Brown University</i>                      |                         |
| <b>PHYS0030 Introductory Physics Workshop Assistant</b>          | Fall 2014               |
| <i>Physics Department, Brown University</i>                      |                         |

## LEADERSHIP & SERVICE

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| <b>Leader &amp; Mentor to Astronomy Graduate Students</b>          | 2023 - present |
| <i>Astronomy Mentorship Program for Upcoming Postdocs (AMP-UP)</i> |                |
| <b>Subject-matter expert reviewer in NASA peer review</b>          | 2025           |
| <i>NASA</i>                                                        |                |
| <b>Pi Day Outreach Organizer</b>                                   | 2024 - 2025    |
| <i>Harvard University</i>                                          |                |

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| <b>Graduate Women in Physics Leader</b><br><i>Massachusetts Institute of Technology</i>                                    | 2019 - 2023    |
| <b>High Performance Computing Mentor</b><br><i>International High Performance Computing Summer School, Athens, Greece</i>  | June 2022      |
| <b>Reviewer for JOSS</b><br><i>Journal of Open Source Software</i>                                                         | 2022 - present |
| <b>President, MIT Salsa Club</b><br><i>Massachusetts Institute of Technology</i>                                           | 2018 - 2022    |
| <b>Physics Representative, Diversity and Inclusion Committee</b><br><i>MIT Graduate Student Council</i>                    | 2018 - 2022    |
| <b>Treasurer, Students for the Exploration and Development of Space</b><br><i>Massachusetts Institute of Technology</i>    | 2017 - 2021    |
| <b>Adopt-a-Physicist Volunteer</b><br><i>Sigma Pi Sigma</i>                                                                | 2018 - 2020    |
| <b>Observer &amp; Designer, MIT Sidewalk Astrogazers</b><br><i>MIT Kavli Institute for Astrophysics and Space Research</i> | 2017 - 2019    |
| <b>Head of Design Team, The Triple Helix Magazine</b><br><i>Brown University</i>                                           | 2014 - 2017    |
| <b>Women in Physics Co-coordinator</b><br><i>Brown University</i>                                                          | 2016 - 2017    |
| <b>Physics Show Presenter</b><br><i>Physics &amp; Astronomy Department, Texas A&amp;M University</i>                       | Summer 2014    |